Open-Source RAG System — Evaluation Report

RBI Document QA · qwen2.5:7b · BAAI/bge-base-en-v1.5 · BAAI/bge-reranker-base · Local Ollama

1. System Overview

Retrieval-Augmented Generation pipeline over RBI Master Directions corpus. Documents chunked (512 tokens, 64 overlap), indexed with FAISS + BGE embeddings. Optional cross-encoder reranking (bge-reranker-base). Generation and LLM judge use local qwen2.5:7b via Ollama. Evaluations run on 40-question benchmark dataset (26 retrieval-labeled, 8 unanswerable, 6 ambiguous).

2. Retrieval: Baseline vs. Reranking (n=40, same dataset)

Configuration	Recall@5	Recall@10	MRR@10	Retrieval p50	Retrieval p95
Baseline (FAISS only)	0.769	0.885	0.648	—	—
+ bge-reranker-base (×4 candidates)	0.962 (+25pp)	0.962 (+8pp)	0.862 (+33pp)	14.1 s	15.2 s

Source: results/reranking.json (retrieval_only=True, 40 Qs). Reranking retrieves 4× candidates then re-scores with cross-encoder; final top-10 returned.

3. Answer Quality — Full-RAG with LLM Judge (n=40)

Metric	Value	Notes
LLM judge mean correctness	0.765	34 judged (6 skipped: ambiguous, no reference answer)
LLM judge mean faithfulness	0.894	Proportion of generated factual claims verifiable in retrieved context
Exact-substring correctness	0.115	Naive baseline heuristic; prefer judge score for semantic correctness
Judge API failures	0	0/34 local Ollama calls failed; all valid JSON returned

Source: results/baseline_qwen.json (full_rag, qwen2.5:7b judge).

4. Abstention & Refusal Detection

Run	Abstention Recall	Abstention Precision	Hallucination Rate	n
Targeted (unanswerable-only)	1.000 (8/8)	1.000 (0 FP)	0.000	8
Full 40-q run (corrected detector)	1.000 (8/8)	0.889 (1 FP)	0.000	40
Full 40-q run (original detector)	0.000 (0/8)	—	1.000	40

Original run: model produced valid refusals using lexical variants ('does not contain data', etc.) that escaped 9-pattern detector. Fix: 5 additional patterns added to is_refusal_response() in generation.py; 8/8 unanswerable detected. 1 false positive in full run.

5. LLM Judge Validation & Human Annotation Status

Validation Dimension	Status / Result	Methodology & Verification Details
LLM Judge Execution	34 judged / 6 skipped	0 failures. Q35–Q40 skipped by design (ambiguous type, empty gold answer).
Judge Bias & Failure Modes	Identified & Verified	Self-evaluation bias (same model qwen2.5:7b). Judge failed on 6/8 unanswerable questions due to distractor context attention drift.
Human-Labelled Samples (20)	Template Prepared	Stratified 20-sample template generated: eval/human_validation_20_template.csv (7 factual, 3 multi-hop, 3 table, 4 unanswerable, 3 ambig).
Cohen's Kappa (Human vs Judge)	Pending Human Labels	Dataset 'answerable' flag is not human rating. Genuine kappa requires human scoring. Automated script: scripts/calculate_kappa.py.

6. Latency Breakdown (Measured, n=40 per row)

Component	mean	p50	p95	min	max	Source
Retrieval + reranking (baseline)	14.0 s	14.1 s	15.2 s	12.2 s	15.6 s	reranking.json
Retrieval + reranking (rerank run)	18.6 s	16.7 s	30.4 s	14.5 s	37.0 s	reranking_qwen.json
Full RAG baseline (no judge)	86.8 s	83.4 s	121.7 s	12.1 s	136.6 s	baseline.json
Full RAG + LLM judge	102.5 s	91.3 s	174.7 s	79.3 s	208.5 s	baseline_qwen.json

7. Best Validated Configuration

Dimension	Best Result	Configuration
Retrieval	Recall@5 = 0.962, MRR@10 = 0.862	FAISS + BGE + bge-reranker-base (retrieves 4× top_k candidates)
Answer correctness & faith	Corr = 0.765, Faith = 0.894 (LLM judge)	qwen2.5:7b, temp=0.0, evidence-grounded prompt
Abstention (unanswerable)	Recall = 1.000, Precision = 0.889	Explicit refusal prompt constraint + expanded refusal pattern detector

8. Key Limitations & Methodological Caveats

• Cohen's kappa: Cannot claim κ=1.0 without genuine human annotation. Dataset metadata was previously conflated with human labels; template and calculation script are now prepared for human input.
• Judge failure on negative queries: LLM judge failed on 6/8 valid refusals because retrieved distractor context distracted the judge prompt.
• Self-evaluation bias: Both generation and LLM judge use qwen2.5:7b, inflating apparent alignment; external oracle recommended.
• Hardware latency: Generation requires 83–102 s per question on CPU/local hardware; requires streaming or GPU acceleration for interactive demo.

Tests: 63/63 pass (pytest -q) · Demo: streamlit run app.py · Human Validation: python scripts/calculate_kappa.py